Sound: pitch and amplitude

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a longitudinal-wave model to distinguish pitch and sound level.

SUCCESS CRITERIA - I can...

1. Describe compressions and rarefactions.
2. Connect pitch primarily with frequency.
3. Separate particle motion from sound propagation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A wave enters a medium where its speed is 3 times as large, at a small angle below critical conditions. Does the transmitted ray bend toward or away from the normal?

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: T = 1/f; v = fλ.
Higher f generally means higher pitch. Amplitude affects intensity under matched conditions, not frequency.

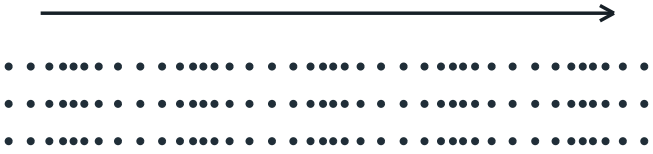
Worked reasoning: A tuning fork makes 440 cycles each second, so its frequency is 440 Hz. A louder version can have the same frequency but greater amplitude.

Your turn: Draw two displacement-time traces with equal periods and different amplitudes.

Marzano check - circle: ① ② ③

03 Look and explain

Mark compressions and rarefactions in the particle picture. Do air particles travel from speaker to listener with the disturbance?



Label dense and sparse regions.

Marzano check - circle: ① ② ③

04 Build the solution

Tone A is 200 Hz; B is 400 Hz. Which has higher pitch? If both play for 2 s, how many cycles does each produce?

Marzano check - circle: ① ② ③

Sound: pitch and amplitude

antocciphysics

05 Use the idea

Turning up a speaker’s volume does not normally double a song’s pitch. Explain which wave property mainly changes.

Marzano check - circle: 1 2 3

06 Choose a method

A refrigerator uses 24 J electrical work while moving heat from its colder interior to a warmer room. Explain why this is not spontaneous cold-to-hot heat flow.

Marzano check - circle: 1 2 3

07 Exit check

Is sound in air mainly transverse or longitudinal? State how particles move relative to propagation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Design two sound changes: one changes pitch only and one changes amplitude only. Describe how their graphs differ.

Marzano check - circle: 1 2 3